6CCSACCM

Cooperation, Coordination and Multi-Agent Systems

Revision Notes

Academic Year 2025/26 — Semester 2

Compiled on January 19, 2026

These notes are structured to help you prepare for exams.
Each week includes: learning objectives, core concepts, theory, and worked examples.

Contents

Module Overview **3**

1 Week 1: Non-Cooperative Games (Part I) **5**

1.1 Topic Summary . . . 5

1.2 Overview + Core Concepts + Extended Theory . . . 6

1.2.1 Multi-Agent Systems . . . 6

1.2.2 Normal Form Games . . . 6

1.2.3 The Prisoner's Dilemma . . . 7

1.2.4 Dominance . . . 8

1.2.5 Iterated Elimination of Dominated Strategies (IEDS) . . . 8

1.2.6 Nash Equilibrium (Pure Strategies) . . . 9

1.2.7 Coordination Games . . . 10

1.2.8 Mixed Strategies . . . 10

1.2.9 Computing Mixed Strategy Nash Equilibria . . . 11

1.2.10 Maximin and Minimax Strategies . . . 12

1.2.11 Zero-Sum Games and the Minimax Theorem . . . 13

1.2.12 Other Solution Concepts . . . 14

1.2.13 Summary: Classic Games . . . 14

1.2.14 Worked Examples from Exercises . . . 15

1.2.15 Other Game Types (Preview) . . . 15

Module Overview

About This Module

This module focuses on **multi-agent systems**—systems consisting of multiple autonomous entities with distributed information, computational capabilities, and potentially conflicting objectives. These systems can be:

- **Cooperative**: sensor networks, teams of mobile robots in a warehouse
- **Competitive**: electronic marketplaces, resource/task allocation
- **Mixed**: collaborative platforms like Wikipedia or Stack Overflow

Agents may be entirely artificial, entirely human, or a mix of both.

Module code: 6CCSACCM **Semester:** 2 (AY 2025/26)

Module Aims

The module aims to:

- Discuss foundational and recent research on various aspects of **multi-agent systems**.
- Place particular emphasis on **coordination strategies** and **fostering cooperative behaviour**.
- Focus on **networks** and the **internet economy**.
- Introduce students to the field and help identify **potential research directions**.
- Develop skills in **critical analysis of research papers** and **formulating research questions**.

Learning Outcomes

By the end of this module, you will be able to:

Code	Learning Outcome
CCM1	Use **game-theoretic, algorithmic and computational tools** to model and analyse multi-agent strategic behaviour and its outcomes.
CCM2	Use **game-theoretic tools** to foster cooperation and stability in multi-agent systems.
CCM3	Use **algorithmic mechanism design techniques** to model and design incentives in multi-agent interaction.
CCM4	Apply **computational social choice techniques** to analyse multi-agent decision-making and improve the quality and stability of collective decisions.

Key Topics (Expected)

Based on the learning outcomes, this module is likely to cover:

Area	Topics
Game Theory	Strategic games, Nash equilibrium, coordination games, Prisoner's Dilemma, repeated games
Cooperation	Mechanisms for fostering cooperation, stability concepts, coalition formation
Mechanism Design	Incentive design, auctions, voting mechanisms, truthfulness
Social Choice	Voting rules, Arrow's theorem, manipulation, collective decision-making
Networks	Network effects, internet economy, platform design

Assessment and Feedback

Formative assessment opportunities:

- **Discussion exercises**: peer discussion and feedback
- **Tutorial questions**: regular problem sheets with feedback during sessions
- **Discussion forums**: post questions on KEATS
- **Office hours**: direct feedback opportunities

[TODO] Add assessment structure (exam format, coursework) when details are provided.

---

1 Week 1: Non-Cooperative Games (Part I)

1.1 Topic Summary

Learning Objectives

By the end of this week, you will be able to:

1. Define the core concepts of non-cooperative game theory, including **strategy profiles**, **payoff functions**, and the distinction between cooperative and non-cooperative games.
2. Formulate real-world scenarios as **strategic-form (normal-form)** or extensive-form games involving rational, self-interested agents.
3. Identify and classify various types of strategies: **pure**, **mixed**, **dominant**, and **best response** strategies.
4. Analyse game-theoretic models to determine the existence and properties of **Nash equilibria** (in pure and mixed strategies).
5. Explain and interpret equilibrium behaviour in classical games: **Prisoner's Dilemma**, **Matching Pennies**, **Battle of the Sexes**, **Stag Hunt**, and coordination games.
6. Evaluate the potential for and limitations of non-cooperative solutions, including issues of **uniqueness**, **multiplicity**, and **inefficiency** of equilibria.
7. Demonstrate competency in using mathematical notation and reasoning to represent and solve strategic games.

The Big Picture

This week introduces **non-cooperative game theory**—the mathematical framework for analysing strategic interactions between rational decision-makers. The module sits at the intersection of two fields:

- **Algorithmic Game Theory (AGT):** Studies algorithms in strategic environments where inputs come from self-interested agents. Combines theoretical CS and economics to analyse equilibria and design efficient, fair mechanisms.
- **Computational Social Choice (COMSOC):** Studies collective decision-making and preference aggregation, asking how individual preferences can be combined into collective decisions.

The central question: *"What is a stable outcome when multiple self-interested agents interact?"*

What Examiners Like to Ask

- Define a game in **normal form**; write the formal tuple $\Gamma = (N, (S_i), (U_i))$.
- Given a payoff matrix, identify **dominated strategies** and solve by iterated elimination.
- Find all **pure strategy Nash equilibria** using best-response analysis.
- Compute **mixed strategy Nash equilibria** using the indifference principle.
- Compare **maximin** vs **minimax** strategies; apply the Minimax theorem to zero-sum games.
- Explain why Prisoner's Dilemma has a unique NE that is Pareto-inefficient.
- Distinguish **Nash equilibrium** from **strong equilibrium**.

1.2 Overview + Core Concepts + Extended Theory

Roadmap

1. Multi-agent systems and game types
2. Normal form games: formal definition
3. Dominance and iterated elimination
4. Nash equilibrium in pure strategies
5. Mixed strategies and expected payoffs
6. Nash equilibrium in mixed strategies
7. Computing mixed NE (indifference principle)
8. Maximin and minimax strategies
9. Zero-sum games and the Minimax theorem
10. Other solution concepts (strong equilibrium, correlated equilibrium)

1.2.1 Multi-Agent Systems

Multi-Agent System (MAS)

A **multi-agent system** consists of multiple autonomous entities (agents) with:

- Distributed information
- Computational capabilities
- Potentially conflicting objectives

Types:

Type	Examples
Cooperative	Sensor networks, robot teams in warehouses
Competitive	Electronic marketplaces, resource allocation
Mixed	Wikipedia, Stack Overflow

1.2.2 Normal Form Games

Non-Cooperative Game

In a **non-cooperative** (or competitive) game:

- Players take actions and enjoy outcomes *individually*
- There are **no binding contracts** among players
- Each player strategically optimises their own utility function

Normal Form Game (Formal Definition)

A game in **strategic (normal) form** is a tuple:

$$\Gamma = (N, (S_i)_{i \in N}, (U_i)_{i \in N})$$

where:

- N is a finite set of **players**
- S_i is the set of **strategies** for player $i \in N$
- $U_i : S \to \mathbb{R}$ is the **utility function** for player i, where $S = \times_{i \in N} S_i$ is the set of all strategy profiles

Strategy Profile Notation

A **profile** s is a vector of strategies $(s_1, \ldots, s_n)$ where $s_i \in S_i$.

Key notation:

- $s_{-i} = (s_1, \ldots, s_{i-1}, s_{i+1}, \ldots, s_n)$ — profile without player i's strategy
- $s = (s_i, s_{-i})$ — decomposition of profile
- s_C — strategies of coalition $C \subseteq N$
- s_{-C} — strategies of players not in C

Exam Tip: Writing Game Definitions

When asked to "formulate as a game", always specify:

1. The set of players N
2. The strategy set S_i for each player
3. The utility function U_i (or payoff matrix for finite games)

1.2.3 The Prisoner's Dilemma

Prisoner's Dilemma

Setup: Two suspects are interrogated in separate rooms. Each can:

- **Cooperate (C):** Deny the crime
- **Defect (D):** Confess (rat out the other)

Payoff matrix:

	C	D
C	$-1, -1$	$-10, 0$
D	$0, -10$	$-5, -5$

Reading the matrix: First number = row player's payoff; second = column player's.

Key insight: The only stable outcome is (D, D), even though (C, C) gives both players a better payoff!

Prisoner's Dilemma: General Form

	C	D
C	R, R	S, T
D	T, S	P, P

Condition: $T > R > P > S$

- T = Temptation (defect while other cooperates)
- R = Reward (mutual cooperation)
- P = Punishment (mutual defection)
- S = Sucker's payoff (cooperate while other defects)

1.2.4 Dominance

Strict and Weak Dominance

For player i and strategies $s, t \in S_i$:

Strict dominance: s ***strictly dominates*** t if:

$$\forall s_{-i} \in S_{-i} : \quad U_i(s, s_{-i}) > U_i(t, s_{-i})$$

Weak dominance: s ***weakly dominates*** t if:

$$\forall s_{-i} \in S_{-i} : U_i(s, s_{-i}) \geq U_i(t, s_{-i}) \quad \text{and} \quad \exists s'_{-i} : U_i(s, s'_{-i}) > U_i(t, s'_{-i})$$

Dominant and Dominated Strategies

- Strategy s is **dominant** if it dominates *all* other strategies
- Strategy t is **dominated** if *some* other strategy dominates it

Key principle: Rational players never choose strictly dominated strategies.

Exam Tip: Dominance is Rare but Powerful

Dominant strategy is a very strong solution concept—no assumptions needed about other players. However, it rarely exists. In Prisoner's Dilemma, Defect is dominant for both players.

1.2.5 Iterated Elimination of Dominated Strategies (IEDS)

IEDS Algorithm

Procedure:

1. Find a strictly dominated strategy for any player
2. Eliminate it from the game
3. Repeat until no dominated strategies remain

Properties:

- **Strict dominance:** Order-independent (same final result)
- **Weak dominance:** Order-dependent (different elimination orders may give different results)
- Surviving strategies need not be dominant
- May end with multiple strategies per player

Example: IEDS on a 3×3 Game

Original game:

	L	C	R
T	$73, 25$	$57, 42$	$66, 32$
M	$80, 26$	$35, 12$	$32, 54$
B	$28, 27$	$63, 31$	$54, 29$

Step 1: For column player, R dominates L (compare: $32 > 25$, $54 > 26$, $29 > 27$). Eliminate L.

Step 2: For row player, T and B both dominate M. Eliminate M.

Step 3: For column player, C dominates R. Eliminate R.

Step 4: For row player, B dominates T ($63 > 57$). Eliminate T.

Solution: (B, C) with payoffs $(63, 31)$.

Verify NE: Row player: $63 > 54$ (B vs T against C). Column player: $31 > 29$ (C vs R against B). ✓

1.2.6 Nash Equilibrium (Pure Strategies)

Nash Equilibrium

A strategy profile $s^* = (s_1^*, \ldots, s_n^*)$ is a **(pure strategy) Nash equilibrium** if no player has a **unilateral improving move**:

$$\forall i \in N, \forall s_i' \in S_i : \quad U_i(s_i^*, s_{-i}^*) \geq U_i(s_i', s_{-i}^*)$$

Equivalently: each player plays a **best response** to the others' strategies.

Best Response

Strategy s_i is a **best response** to s_{-i} if:

$$s_i \in \arg\max_{s_i' \in S_i} U_i(s_i', s_{-i})$$

Exam Tip: Finding Pure Strategy NE

Method 1 (Best Response):

1. For each opponent strategy, find player's best response (underline/mark it)
2. NE = cells where *all* players are playing best responses

Method 2 (Check deviations):

1. For each cell, check if any player wants to deviate unilaterally
2. NE = cells with no profitable deviations

Example: Best Response Analysis

	L	C	R
T	$2, 2$	$1^*, 3^*$	$0^*, 1$
M	$3^*, 1^*$	$0, 0$	$0^*, 0$
B	$1, 0^*$	$0, 0^*$	$0^*, 0^*$

Reading: * marks best responses. NE = cells where both payoffs are marked.

Nash equilibria: None! No cell has both payoffs marked.

1.2.7 Coordination Games

Coordination Game

A game where players benefit from **choosing corresponding strategies**:

	Left	Right
Up	$1, 1$	$0, 0$
Down	$0, 0$	$1, 1$

Properties:

- **Two pure strategy NE:** (Up, Left) and (Down, Right)
- No dominant or dominated strategies
- Both NE are also **strong equilibria**

Battle of the Sexes

Players want to coordinate but have **different preferences**:

	Theater	Football
Theater	$2, 1$	$0, 0$
Football	$0, 0$	$1, 2$

Pure NE: (Theater, Theater) and (Football, Football)

Stag Hunt

	Stag	Hare
Stag	$4, 4$	$1, 3$
Hare	$3, 1$	$3, 3$

Story: Two hunters can cooperate for a stag (high reward) or safely catch a hare alone.

Pure NE: (Stag, Stag) and (Hare, Hare)—plus mixed equilibria (see below).

1.2.8 Mixed Strategies

Mixed Strategy

A **mixed strategy** is a probability distribution over pure strategies.

- $\Delta_i = \Delta(S_i)$ = set of probability distributions over S_i
- A pure strategy is a mixed strategy with probability 1 on one action
- Even with finite S_i, there are infinitely many mixed strategies

Expected Utility

When mixed strategy profile $\tilde{s} \in \Delta$ is played:

$$U_i(\tilde{s}) = \sum_{s \in S} U_i(s) \cdot \Pr(s|\tilde{s})$$

where:

$$\Pr(s|\tilde{s}) = \prod_{j \in N} \tilde{s}_j(s_j)$$

(product of individual probabilities, assuming independence)

Nash's Existence Theorem (1951)

Every finite game (finite players, finite action sets) has at least one **mixed strategy Nash equilibrium**.

1.2.9 Computing Mixed Strategy Nash Equilibria

Indifference Principle

In a mixed strategy NE where player i randomises over actions $A \subseteq S_i$:

Player i must be indifferent among all actions in A:

$$\forall a, a' \in A: \quad U_i(a, \tilde{s}_{-i}) = U_i(a', \tilde{s}_{-i})$$

Key insight: The *other* player's mixing probabilities make you indifferent!

Example: Matching Pennies

	H	T
H	$1, -1$	$-1, 1$
T	$-1, 1$	$1, -1$

No pure strategy NE (verify: every cell has a profitable deviation).

Mixed NE: Let Player 1 play H with probability p, Player 2 play H with probability q.

Player 2's indifference:

$$U_2(\text{H}) = U_2(\text{T}) \implies -p + (1-p) = p - (1-p) \implies p = \frac{1}{2}$$

By symmetry: $q = \frac{1}{2}$

Mixed NE: $\left(\frac{1}{2}, \frac{1}{2}\right), \left(\frac{1}{2}, \frac{1}{2}\right)$ with expected utility 0 for both.

Example: Stag Hunt Mixed NE

	Stag	Hare
Stag	$4, 4$	$1, 3$
Hare	$3, 1$	$3, 3$

Let $p = \Pr(\text{Player 1 plays Stag})$, $q = \Pr(\text{Player 2 plays Stag})$.

Player 2's indifference:

$$U_2(\text{Stag}) = U_2(\text{Hare}) \implies 4p + 1(1-p) = 3p + 3(1-p)$$

$$4p + 1 - p = 3p + 3 - 3p \implies 3p + 1 = 3 \implies p = \frac{2}{3}$$

By symmetry: $q = \frac{2}{3}$

Three NE total: Two pure + one mixed at $\left(\frac{2}{3}, \frac{2}{3}\right)$.

Exam Tip: Mixed NE Computation

1. Set up probability variables (p, q, etc.)
2. Write expected utility equations for each action
3. Set utilities equal (indifference condition)
4. Solve for the *other* player's probability
5. Verify the solution makes sense (probabilities in $[0, 1]$)

Common error: Solving for your own probability instead of the opponent's!

1.2.10 Maximin and Minimax Strategies

Maximin (Safety Level) Strategy

The **maximin strategy** for player i maximises their worst-case payoff:

$$s_i^{LS} \in \arg\max_{s_i} \min_{s_{-i}} U_i(s_i, s_{-i})$$

The **maximin value** (safety level) is:

$$U_i^{LS} = \max_{s_i} \min_{s_{-i}} U_i(s_i, s_{-i})$$

Intuition: If opponents play to hurt you as much as possible, what's your best guarantee?

Minimax Strategy (2-Player Games)

The **minimax strategy** for player i against player $-i$ minimises $-i$'s maximum payoff:

$$s_i^{MM} \in \arg\min_{s_i} \max_{s_{-i}} U_{-i}(s_i, s_{-i})$$

Intuition: How can you punish the opponent most, ignoring your own payoff?

Maximin in Prisoner's Dilemma

	C	D
C	$-1, -1$	$-10, 0$
D	$0, -10$	$-5, -5$

Row player's analysis:

- If commit to C: opponent plays D (minimises row), payoff $= -10$
- If commit to D: opponent plays D (minimises row), payoff $= -5$

Maximin: Choose D (guarantees $-5 > -10$)

Defect is both the dominant strategy AND the maximin strategy.

1.2.11 Zero-Sum Games and the Minimax Theorem

Constant-Sum and Zero-Sum Games

A two-player game is **constant-sum** if $\exists c$ such that:

$$\forall s \in S_1 \times S_2 : \quad U_1(s) + U_2(s) = c$$

For $c = 0$: **zero-sum game** (pure conflict—one player's gain is the other's loss).

Minimax Theorem (von Neumann, 1928)

For any finite, two-player, zero-sum game:

$$\max_{s_1} \min_{s_2} U_1(s_1, s_2) = \min_{s_2} \max_{s_1} U_1(s_1, s_2)$$

Consequences:

1. Each player's **maximin value = minimax value**
2. The common value is called the **value of the game**
3. **Maximin strategies = minimax strategies = NE strategies**
4. All Nash equilibria have the **same payoff vector**

Matching Pennies: Minimax Analysis

	H	T
H	$1, -1$	$-1, 1$
T	$-1, 1$	$1, -1$

This is a zero-sum game ($U_1 + U_2 = 0$ always).

Value of the game: 0

By Minimax theorem: The unique NE $\left(\frac{1}{2}, \frac{1}{2}\right)$ is also the maximin = minimax strategy for each player.

Computing Maximin/Minimax in General-Sum Games

For a general-sum 2-player game:

1. Player i's maximin depends *only* on i's utilities
2. Create a zero-sum game: keep i's utilities, negate them for $-i$
3. Solve the zero-sum game for NE
4. NE strategies = maximin strategies

1.2.12 Other Solution Concepts

Strong Equilibrium

A profile s is a **strong equilibrium** if no coalition $C \subseteq N$ has a joint improving move:

$$\nexists C \subseteq N, s'_C \in S_C : \quad \forall i \in C : U_i(s'_C, s_{-C}) > U_i(s)$$

Key facts:

- Every strong equilibrium is a Nash equilibrium (consider singleton coalitions)
- Much more rare than NE
- Prisoner's Dilemma has **no** strong equilibrium
- Coordination game: both pure NE are strong equilibria

Correlated Equilibrium

Players condition actions on a shared random signal from a trusted mediator.

- Every mixed NE induces a correlated equilibrium
- Set of correlated equilibria $\supseteq$ set of NE
- **Computationally easier:** Can be found via linear programming

ε-Nash Equilibrium

A profile where no player can gain more than $\varepsilon > 0$ by deviating:

$$\forall i, \forall s'_i : \quad U_i(s_i, s_{-i}) \geq U_i(s'_i, s_{-i}) - \varepsilon$$

Standard NE is the special case $\varepsilon = 0$.

Rationalizable Strategies

A strategy is **rationalizable** if it is a best response to some belief about opponents' strategies (consistent with common knowledge of rationality).

- Equivalent to surviving iterated elimination of never-best-responses
- Every NE strategy is rationalizable
- Set of rationalizable strategies $\supseteq$ NE strategies

1.2.13 Summary: Classic Games

Game	Pure NE	Mixed NE	Strong NE	Key Feature
Prisoner's Dilemma	1 (D,D)	Same	None	Dominant strategy, Pareto-inefficient
Coordination	2	+1	2	Multiple equilibria
Matching Pennies	0	1	0	Zero-sum, no pure NE
Battle of Sexes	2	+1	2	Asymmetric preferences
Stag Hunt	2	+1	2	Risk vs reward trade-off

1.2.14 Worked Examples from Exercises

Exercise: Dividing Money

Setup: Two players divide £10. Each names $x_i \in \{0, 1, \ldots, 10\}$.

- If $x_1 + x_2 \leq 10$: each gets their claim (remainder destroyed)
- If $x_1 + x_2 > 10$ and $x_1 \neq x_2$: lower claimant gets x_i, other gets rest
- If $x_1 + x_2 > 10$ and $x_1 = x_2$: each gets £5

Best responses:

Other's claim	Best responses
0	$\{10\}$
1	$\{9, 10\}$
2	$\{8, 9, 10\}$
$\vdots$	$\vdots$
5	$\{5, 6, 7, 8, 9, 10\}$
6	$\{5, 6\}$
7	$\{6\}$
8	$\{7\}$
9	$\{8\}$
10	$\{9\}$

Nash equilibria: $(5, 5)$, $(5, 6)$, $(6, 5)$, $(6, 6)$ — any pair where both are best responses to each other.

Exercise: Modified Stag Hunt

Modified payoffs:

	Stag	Hare
Stag	$4, 4$	$1, 3$
Hare	$3, 2$	$3, 3$

Whose mixing probability changes?

Player 2's utility changed ($1 \to 2$ in Hare-Stag cell).

Effect: Player 2's utility is used to determine Player 1's indifference condition.

Player 1's mixing probability changes: $p = \frac{1}{2}$ (was $\frac{2}{3}$).

Player 2's mixing probability stays: $q = \frac{2}{3}$ (Player 1's utilities unchanged).

1.2.15 Other Game Types (Preview)

Game Type	Description
Extensive form	Game tree; players move in turns (sequential)
Repeated games	Same game played multiple times; cumulative payoffs
Cooperative (coalitional)	Players form coalitions; binding contracts; characteristic function over coalitions

These will be covered in subsequent weeks.